

Accessing Quantum History in the New Century: A Collaborative (Re)Vision of Archival Governance

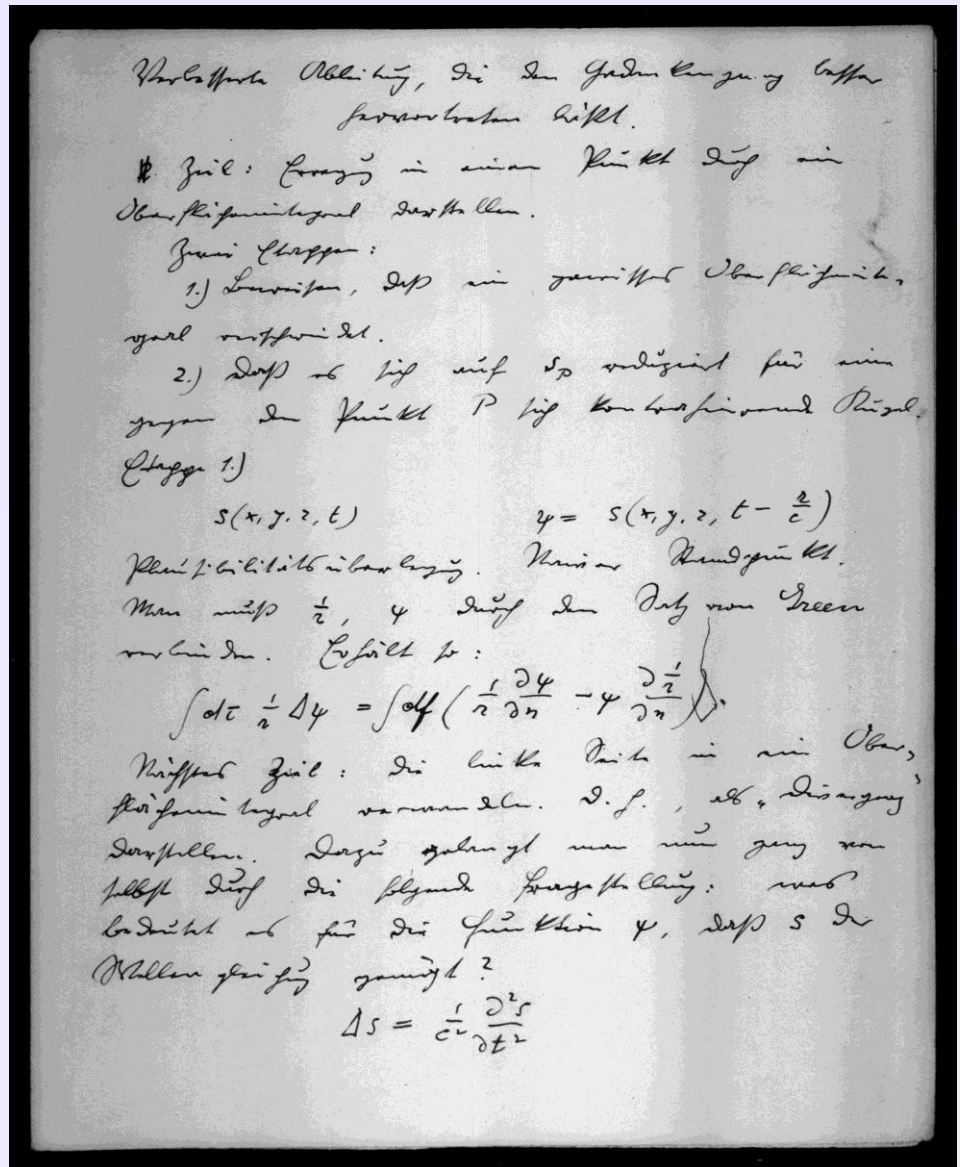
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“Never in the history of science has so effective an effort been made to record decisive moments in the evolution of new ideas while key participants are still alive.”^[1]

Background

Sixty years after Max Planck introduced quantum theory, scientists and historians increasingly recognized that the rapid development of modern physics had outpaced efforts to document its history. Declining personal correspondence, more impersonal scientific publishing, and the growing technical complexity of the field made it difficult to reconstruct the origins of major ideas, while unpublished sources such as notebooks, letters, and reminiscences remained



Digitized microfilm copy of notes by Erwin Schrödinger developing his theory of waves, circa 1926.^{[2][3]}

essential. In response, the Archive for the History of Quantum Physics (AHQP) was established in 1960 by the Joint Committee of the American Physical Society and the American Philosophical Society, with initial National Science Foundation (USA) funding obtained by AIP. The project compiled manuscript material, recorded oral histories, published a research catalog, and distributed original or copied materials to major research libraries worldwide. Designed as an international, multi-institutional initiative, the AHQP drew contribution from scientific institutions including the Niels Bohr Archive (Denmark) and the Max Planck Institute (Germany), while libraries such as the National Diet Library (Japan) and the University of Melbourne (Australia) became “Libraries of Deposit” providing access to microfilm copies of the collection.^[1]

Problem

Archival practice and technology have changed substantially since the AHQP was established. Its original access protocols balanced scholarly use with the literary rights of manuscript creators and oral history participants by limiting access, restricting quotation and reproduction, and controlling distribution. Although appropriate at the time, these requirements have increasingly hindered access, long-term stewardship, and use of the AHQP as an integrated resource, particularly by limiting digital access and preservation initiatives sought by the broader history of science community.

Solution



Left to right: Otto Hahn, Werner Heisenberg, Lise Meitner, and Max Born at Lindau Conference of Nobel Laureates, 1962.^[4]

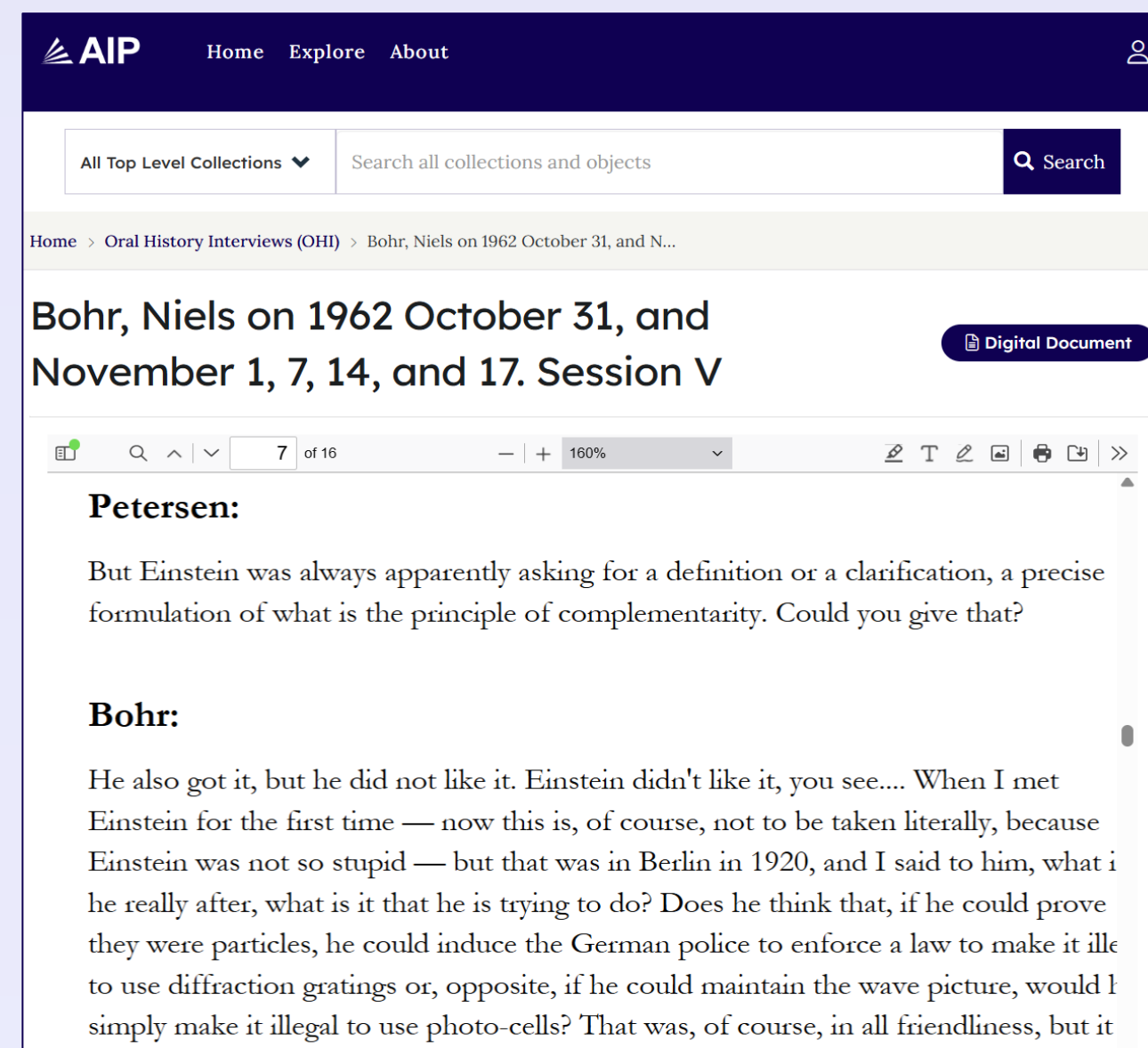
Catalyzed by research, conversations, and celebrations during UNESCO’s 2025 International Year of Quantum Science & Technology, the American Philosophical Society and AIP, with feedback from other partner institutions such as the Niels Bohr Archive, have collaborated to address these challenges by developing a new convention document based on open-access principles.

Core Drafted Revisions

- Stating the AHQP partner institution’s commitment to open access.
- Removing language that overly restricts use.
- Granting AHQP managing institutions equal capacity to authorize use.
- Leaving flexibility for future archival growth and adaptation.

Community

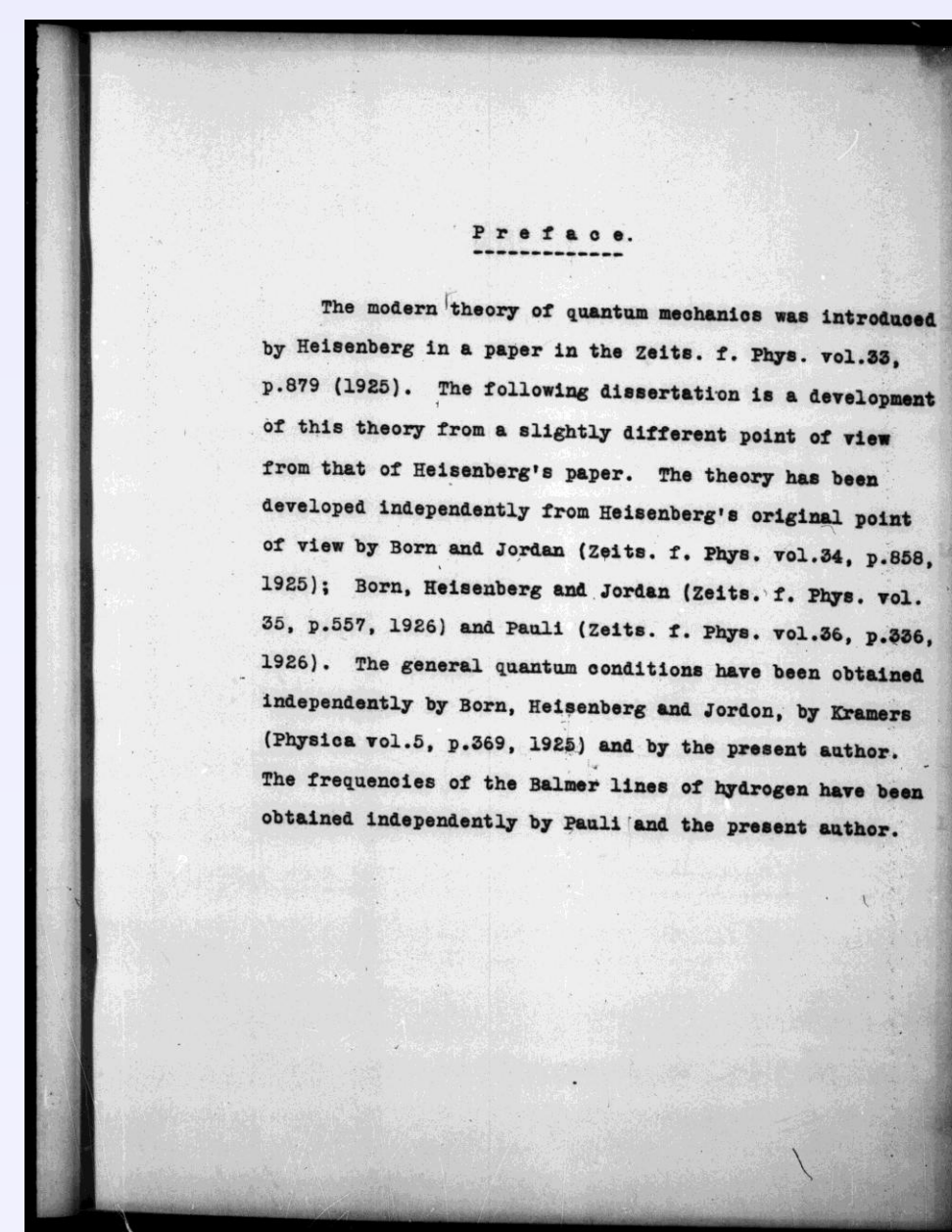
The revised policy seeks to balance responsible rights management with expanded international access, visibility, and research use of the AHQP for scholarly and public



Computer generated transcript of an oral history interview with Niels Bohr, recorded the day before Bohr’s death in Copenhagen, Denmark.^[5]

audiences. A draft agreement has been completed and is undergoing final review. Multi-institutional collaboration remains central: Libraries of Deposit holding AHQP microfilm were contacted, and respondents expressed strong support for clearer governance, having faced similar uncertainty over access and permissions under the original convention. Some institutions have digitized sections of their holdings but limited online access because of these ambiguities. Once finalized, the revised agreement will provide stakeholders with a clearer, more open framework for administering this international, multi-institutional archive.

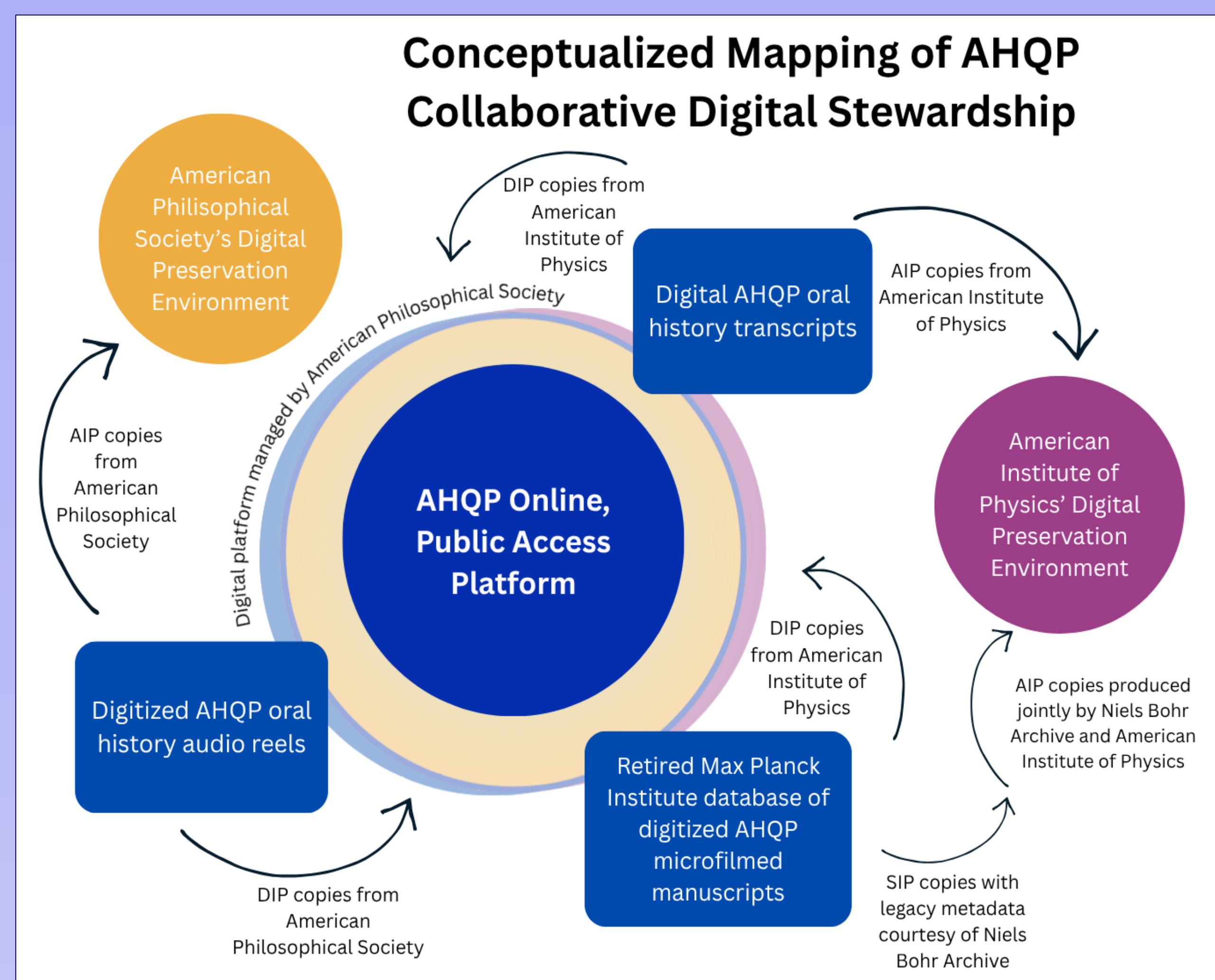
Preservation & Horizon



Digitized microfilm copy Paul Dirac’s 1926 PhD dissertation, the first doctoral thesis written on the subject of quantum mechanics.^[6]

Collaborating on a renewed vision and shared governance model for the AHQP will enable participating institutions to improve digital preservation and access. Recent initiatives demonstrate the potential of this partnership. The American Philosophical Society received a CLIR *Recordings at Risk* grant to digitize the AHQP oral history recordings held on deteriorating reels. The Niels Bohr Archive has also begun transferring a database of digitized AHQP microfilm manuscripts to AIP’s preservation environment. Originally developed by the Max Planck Institute for restricted research, the database was later decommissioned and survived only as a hard drive copy held by an NBA affiliate. AIP will preserve the files with the goal of eventually reprocessing and providing access under the revised open-access convention.

To further support preservation and access, the collaborating institutions are developing plans for a unified platform to serve as a single access point to AHQP materials held across repositories. A shared digital stewardship model also creates opportunities to extend documentation of quantum physics beyond the original AHQP. Although active collecting ended in the 1960s, quantum research has continued to develop and is increasingly documented in born-digital formats. A renewed AHQP could therefore support new and expanded collaborations among institutions with shared commitments to collecting, preserving, and providing access to the history of quantum science.



References

- [1] T. S. Kuhn, J. L. Heilbron, and L. Allen, *Sources for History of Quantum Physics*, <https://www.amphilsoc.org/guides/ahqp/index.htm>.
- [2] AHQP, reel 40, section 5, item 3.
- [3] Renn, J. (2013). Schrödinger and the Genesis of Wave Mechanics. *Max Planck Institute for the History of Science Preprint Series*, (437). <https://www.mpiwg-berlin.mpg.de/sites/default/files/s/Preprints/P437.pdf>.
- [4] Laureate Conference. AIP Emilio Segrè Visual Archives. Niels Bohr Library & Archives, AIP, College Park, MD USA. <https://repository.aip.org/node/25499>.
- [5] Interview of Niels Bohr by Thomas S. Kuhn, Leon Rosenfeld, Aage Petersen, and Erik Rudinger on 17 November 1962, AHQP, Niels Bohr Library & Archives, AIP, College Park, MD USA. <https://repository.aip.org/node/127798>.
- [6] AHQP, reel 77.